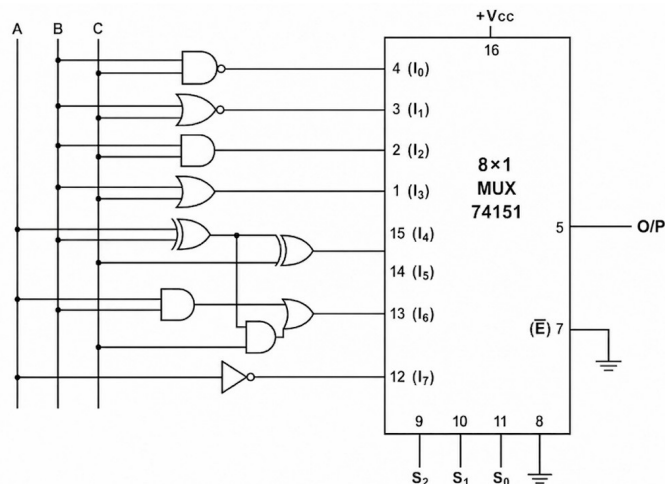


Block diagram:



RTL code with testbench:

```

module alu ( input wire a,
input wire b,
input wire c,
input wire [2:0] s,
output reg o
);

always @(*)begin
case(s)
3'b000: o = ~(b & c);
3'b001: o = ~(b | c);
3'b010: o = (b & c);
3'b011: o = (b | c);
3'b100: o = (b ^ c);
3'b101: o = ((a^b)^c);
3'b110: o = (((a ^ b) & c) | (a & b));
3'b111: o = ~(a);
endcase
end

```

```

`timescale 1ns/1ps
module tb_alu;

reg a;
reg b;
reg c;
reg [2:0] s;
wire o;

alu one (.a(a), .b(b), .c(c), .s(s), .o(o));

initial begin
a = 0; b = 1; c = 1;

$display("A B C | S2 S1 S0 | 0");

for (integer i = 0; i < 8; i = i + 1) begin
s = i[2:0];
#10;
$display("%b %b %b | %b %b %b | %b", a, b, c, s[2], s[1], s[0], o);
end
$finish;
end

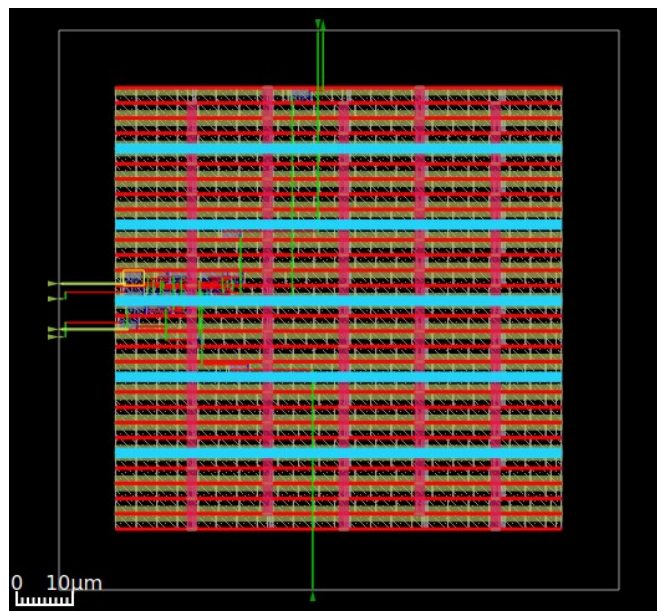
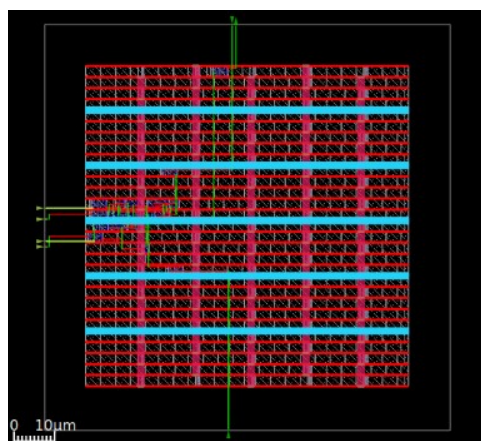
endmodule

```

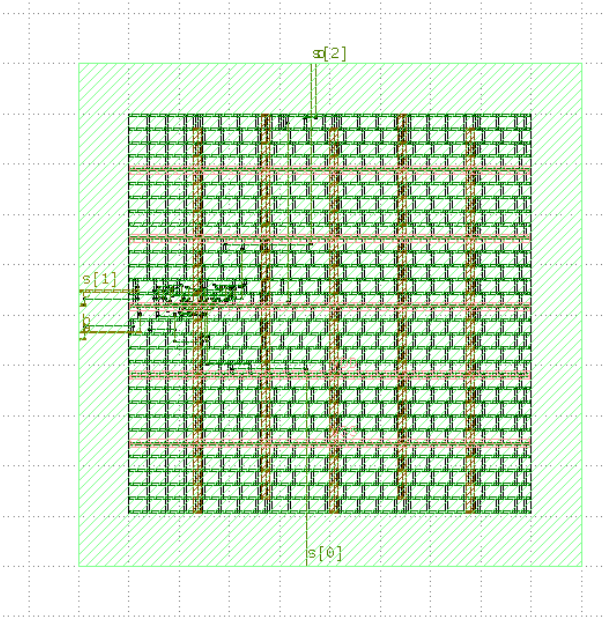
Output:

A	B	C	S2	S1	S0	O
0	1	1	0	0	0	0
0	1	1	0	0	1	0
0	1	1	0	1	0	1
0	1	1	0	1	1	1
0	1	1	1	0	0	0
0	1	1	1	0	1	0
0	1	1	1	1	0	1
0	1	1	1	1	1	1

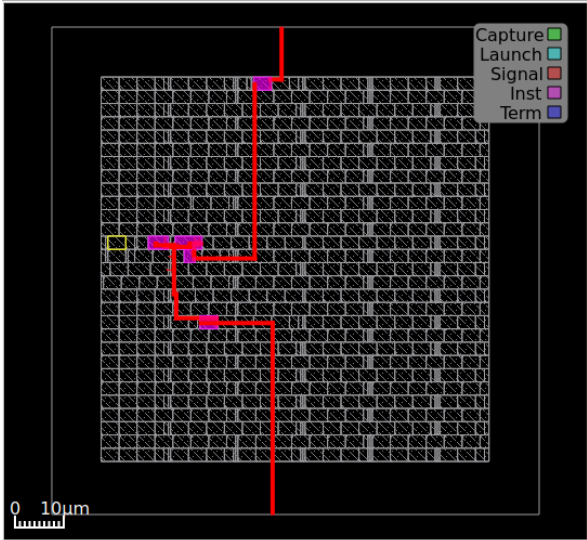
Physical Design:



Klayout view:



An Example Path:

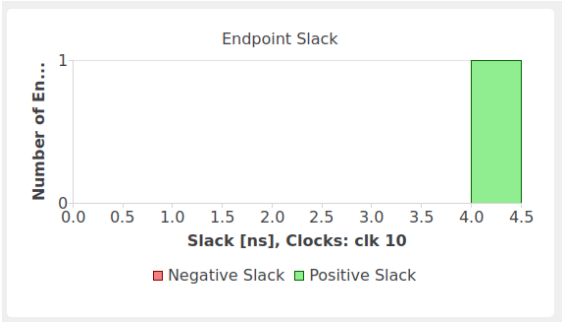


Setup	Hold					
Capture Clock	Required	Arrival	Slack	Skew	Logic	
clk	8.000	3.841	4.159	0.000		
clk	8.000	3.796	4.204	0.000		
clk	8.000	3.769	4.231	0.000		
clk	8.000	3.768	4.232	0.000		
clk	8.000	3.755	4.245	0.000		

Data Path Details	Capture Path Details					
Pin	Fanout	Time	Delay	Slew	Load	
s[0]	1	2.000	2.000	0.000		
clock network delay		2.000	0.000			
input4/A (sky130_fd_sc_hd_clkdlybuf...)		2.000	0.000	0.000		
input4/X (sky130_fd_sc_hd_clkdlybuf...)	4	2.487	0.486	0.142	0.016	

Inspector | Hierarchy Browser | Timing Report | Charts | Help Browser

Timing Analysis:



Setup	Hold					
Capture Clock	Required	Arrival	Slack	Skew	Logic	
clk	8.000	3.841	4.159	0.000		
clk	8.000	3.796	4.204	0.000		
clk	8.000	3.769	4.231	0.000		
clk	8.000	3.768	4.232	0.000		
clk	8.000	3.755	4.245	0.000		

Fanout	Cap	Slew	Delay	Time	Description
		0.00	0.00	0.00	clock vclk (rise edge)
			0.00	0.00	clock network delay (ideal)
			0.20	0.20	v input external delay
4	5.66	0.00	0.00	0.20	v c (in)
2	2.45	0.01	0.05	0.25	v _28_/Z (XOR2_X1)
1	1.57	0.02	0.03	0.28	^ _30_/ZN (OAI21_X1)
1	1.55	0.01	0.02	0.30	v _31_/ZN (OAI21_X1)
1	0.00	0.02	0.03	0.34	^ _32_/ZN (AOI22_X1)
		0.02	0.00	0.34	^ o (out)
				0.34	data arrival time
		0.00	2.00	2.00	clock vclk (rise edge)
			0.00	2.00	clock network delay (ideal)
			0.00	2.00	clock reconvergence pessimism
			-0.20	1.80	output external delay
				1.80	data required time
				1.80	data required time
				-0.34	data arrival time
				1.46	slack (MET)